

Problem 1 :

S.No.	Line No.	Error	Error Type	Error Correction
1	8	Missing semicolon after int x = 10	Syntax Error	Add ; at the end of the statement
2	12	Assignment operator = used in condition if(x = 10)	Logical Error	Use comparison operator ==
3	17	String "Ninety" assigned to integer variable marks	Type Mismatch Error	Assign an integer value
4	19	Variable z used without declaration	Semantic Error	Declare and initialize z before use
5	21	Cannot assign value to constant (100 = x)	Syntax Error	Assign value to a variable instead
6	25	Array index 10 exceeds size of arr[5]	Array Out of Bounds Error	Use index within range 0-4
7	27	Address of integer variable assigned to float *ptr	Type Mismatch Error	Use compatible pointer type
8	29	%f format specifier used with integer variable x	Format Specifier Error	Use %d for integer values
9	34	Division by zero (a/b, where b = 0)	Runtime Error	Ensure divisor is not zero
10	36	Multi-character constant 'AB' assigned to char	Semantic Error	Use a single character constant
11	39	Both if and else print the same message	Logical Error	Print appropriate message in else block
12	44	Function show() called without declaration	Compilation Error	Declare function prototype before use
13	44	Function show() not defined	Linker Error	Provide function definition

S.No.	Line No.	Error	Error Type	Error Correction
14	46	Semicolon after while(x < 15) creates an empty loop	Logical Error	Remove unnecessary semicolon
15	51	continue used outside any loop	Syntax Error	Use continue only inside loops
16	53	break used outside any loop/switch	Syntax Error	Use break only inside loop or switch
17	55	Function calculate() called with one argument instead of two	Function Argument Error	Pass both required arguments
18	34	Program may terminate due to divide-by-zero before later statements execute	Runtime Error	Handle divisor validation before division

Problem 2 :

S.No.	Line No.	Error	Error Type	Error Correction
1	8	Missing semicolon after int num2 = 20	Syntax Error	Add ; at the end of the statement
2	19	Missing closing parenthesis in if(choice == 1	Syntax Error	Add) before {
3	30	Extra closing parenthesis in else if(choice == 3))	Syntax Error	Remove extra)
4	45	Missing closing parenthesis in for(int i=0; i<5; i++	Syntax Error	Add) before {
5	52	Missing closing parenthesis in while(choice > 0	Syntax Error	Add) before {
6	60	Missing semicolon after while(choice > 5) in do-while loop	Syntax Error	Add ; after condition

S.No.	Line No.	Error	Error Type	Error Correction
7	67	Missing colon (:) after case 2	Syntax Error	Add : after case label
8	92	Missing closing quote in char grade = 'A;	Syntax Error	Close the character constant properly
9	96	Missing closing bracket in int arr[5;	Syntax Error	Add] before ;
10	121	Missing semicolon after printf("End of Program\n")	Syntax Error	Add ; at the end
11	102-119	Opening block { is not properly closed	Syntax Error	Add required closing brace }
12	3	Function add() declared but never used	Warning	Call the function or remove declaration
13	38	Integer division performed in num1 / num2	Logical Warning	Use appropriate data type if decimal result is required
14	20-23	Addition performed directly instead of using declared add() function	Logical Warning	Use add() if intended
15	55	While loop only decrements choice and has no visible output	Logical Warning	Add meaningful processing if required
16	57-60	Do-while loop condition may not behave as intended	Logical Error	Verify loop condition according to requirement
17	92	Unterminated character literal causes compilation failure	Compilation Error	Use valid character constant
18	96	Invalid array declaration due to missing bracket	Compilation Error	Complete declaration syntax
19	102-119	Nested block structure becomes unbalanced	Syntax Error	Ensure all opened blocks are properly closed

S.No.	Line No.	Error	Error Type	Error Correction
20	123	Compiler may report unexpected end of file because of unmatched braces	Compilation Error	Match all opening and closing braces

Problem 3 :

S.No.	Line No.	Error	Error Type	Error Correction
1	8	String "Twenty" assigned to integer variable age	Type Mismatch Error	Assign an integer value
2	14	String "A" assigned to char variable grade	Type Mismatch Error	Use a single character constant
3	16	%d format specifier used with float variable salary	Format Specifier Error	Use %f
4	18	%f format specifier used with integer variable marks	Format Specifier Error	Use %d
5	22	Variable score used without declaration	Semantic Error	Declare and initialize score
6	25	Variable x redeclared in the same scope	Compilation Error	Remove duplicate declaration or rename variable
7	27	Cannot assign value to constant (100 = x)	Syntax Error	Assign value to a variable instead
8	31	%s format specifier used with integer variable num	Format Specifier Error	Use %d
9	35	%d format specifier used with float variable pi	Format Specifier Error	Use %f
10	39	Array index 2.5 is not an integer	Array Index Error	Use an integer index

S.No.	Line No.	Error	Error Type	Error Correction
11	41	Float value assigned to char variable ch	Type Conversion Warning	Use valid character data
12	45	Function add() called with one argument instead of two	Function Argument Error	Pass both required arguments
13	51	Function average() called with two arguments instead of three	Function Argument Error	Pass all required arguments
14	55	String "35" assigned to integer variable temperature	Type Mismatch Error	Assign an integer value
15	59	String "10.5" assigned to float variable discount	Type Mismatch Error	Assign a numeric float value
16	67	Direct assignment name = "Robil" to character array	Compilation Error	Use string copy function
17	73	Variable totalMarks used without declaration	Semantic Error	Declare and initialize totalMarks
18	81	%d format specifier used with float variable rate	Format Specifier Error	Use %f
19	89	Variable c used without declaration	Semantic Error	Declare c before use
20	89	Function add() called with three arguments instead of two	Function Argument Error	Pass only two arguments
21	103	Assignment operator used in if(studentMarks = 100)	Logical Error	Use comparison operator ==
22	109	String "Large" assigned to integer variable size	Type Mismatch Error	Assign an integer value
23	115	%d format specifier used with float variable percentage	Format Specifier Error	Use %f
24	119	String "five" used as array index	Array Index Error	Use an integer index

S.No.	Line No.	Error	Error Type	Error Correction
25	128	String constant "2" used in case label of integer switch	Type Mismatch Error	Use integer constant 2
26	136	Function multiply() called without declaration/definition	Compilation / Linker Error	Declare and define the function
27	146	%d format specifier used with float variable areaRect	Format Specifier Error	Use %f
28	154	Return value of average() (float) assigned to integer variable avgMarks	Type Conversion Warning	Use float variable or explicit conversion
29	160	Variable value used without declaration	Semantic Error	Declare and initialize value
30	162	String "EMP101" assigned to integer variable employeeId	Type Mismatch Error	Use a string variable
31	164	%d format specifier used for string-based employee ID	Format Specifier Error	Use %s with string variable

Problem 4 :

S.No.	Line No.	Error	Error Type	Error Correction
1	26	Function calculateSalary() declared but not defined	Linker Error	Provide function definition
2	28	Function printStudentDetails() declared but not defined	Linker Error	Provide function definition
3	34	Function showResult() declared but not defined	Linker Error	Provide function definition
4	36	Function calculateAverage() declared but not defined	Linker Error	Provide function definition

S.No.	Line No.	Error	Error Type	Error Correction
5	44	Function generateReport() called without declaration	Compilation Error	Declare function prototype before use
6	44	Function generateReport() not defined	Linker Error	Provide function definition
7	46	Function displayEmployee() called without declaration	Compilation Error	Declare function prototype before use
8	46	Function displayEmployee() not defined	Linker Error	Provide function definition
9	48	Function processData() called without declaration	Compilation Error	Declare function prototype before use
10	48	Function processData() not defined	Linker Error	Provide function definition
11	50	Function saveRecord() called without declaration	Compilation Error	Declare function prototype before use
12	50	Function saveRecord() not defined	Linker Error	Provide function definition
13	52	Function loadRecord() called without declaration	Compilation Error	Declare function prototype before use
14	52	Function loadRecord() not defined	Linker Error	Provide function definition
15	54	Function printInvoice() called without declaration	Compilation Error	Declare function prototype before use
16	54	Function printInvoice() not defined	Linker Error	Provide function definition
17	56	Function calculateTax() called without declaration	Compilation Error	Declare function prototype before use
18	56	Function calculateTax() not defined	Linker Error	Provide function definition

S.No.	Line No.	Error	Error Type	Error Correction
19	58	Function validateUser() called without declaration	Compilation Error	Declare function prototype before use
20	58	Function validateUser() not defined	Linker Error	Provide function definition
21	60	Function sendNotification() called without declaration	Compilation Error	Declare function prototype before use
22	60	Function sendNotification() not defined	Linker Error	Provide function definition
23	62	Function generateCertificate() called without declaration	Compilation Error	Declare function prototype before use
24	62	Function generateCertificate() not defined	Linker Error	Provide function definition
25	40	companyProfit declared as extern but no definition provided	Linker Error	Define companyProfit globally
26	15	Variable choice declared but never used	Warning	Remove variable or use it in program logic

Problem 5 :

S.No.	Line No.	Error	Error Type	Error Correction
1	11	Division by zero (a / b, where b = 0)	Runtime Error	Ensure divisor is not zero before division
2	15	NULL pointer ptr1 dereferenced	Runtime Error	Assign valid memory before dereferencing
3	19	Array index 10 exceeds size of arr[5]	Array Out of Bounds Error	Use valid indices 0-4
4	23	Uninitialized pointer ptr2 dereferenced	Runtime Error	Initialize pointer before use

S.No.	Line No.	Error	Error Type	Error Correction
5	27	String "Programming" exceeds size of name[5]	Buffer Overflow Error	Increase array size or shorten string
6	33	Uninitialized pointer ptr3 dereferenced	Runtime Error	Initialize pointer before dereferencing
7	39	fscanf() used without checking if file opened successfully	Runtime Error	Verify fp != NULL before reading
8	45	String "Computer Science" exceeds allocated memory of ptr4	Buffer Overflow Error	Allocate sufficient memory
9	51	Accessing memory after free(ptr4)	Use After Free Error	Do not access freed memory
10	57	Double free of ptr5	Runtime Error	Free memory only once
11	63	Modulo by zero (x % y, where y = 0)	Runtime Error	Ensure divisor is not zero
12	69	strcat() causes buffer overflow in str1[10]	Buffer Overflow Error	Ensure destination array has enough space
13	77	Writing 10 elements into memory allocated for only 3 integers	Buffer Overflow Error	Allocate sufficient memory or reduce loop limit
14	85	Accessing matrix[5][5] outside 3×3 array bounds	Array Out of Bounds Error	Use valid row and column indices
15	89	NULL pointer ptr6 dereferenced	Runtime Error	Check pointer before access
16	93	Use of gets() function	Unsafe Function Error	Use a safer input function
17	103	Writing to memory after free(ptr7)	Use After Free Error	Do not access freed memory

S.No.	Line No.	Error	Error Type	Error Correction
18	107	Variable age used without initialization	Uninitialized Variable Error	Initialize variable before use
19	111	Negative array index arr[-1]	Array Out of Bounds Error	Use valid non-negative indices
20	115	Uninitialized pointer ptr8 dereferenced in condition	Runtime Error	Initialize pointer before use
21	123	Printing NULL string pointer text	Runtime Error	Check pointer before printing
22	129	malloc() return value not checked before use	Runtime Error	Verify allocation success before access
23	141	Infinite loop while(1)	Logical Error	Add termination condition
24	39	File "unknown.txt" may not exist	Runtime Error	Check file existence before reading
25	75	Dynamically allocated memory numbers may be corrupted due to overflow	Runtime Error	Stay within allocated bounds
26	127	Extremely large memory allocation may fail	Memory Allocation Error	Validate allocation result
27	51	Dangling pointer used after memory release	Runtime Error	Set pointer to NULL after free()
28	103	Dangling pointer ptr7 used after free()	Runtime Error	Avoid accessing freed memory

Problem 6 :

S.No.	Line No.	Error	Error Type	Error Correction
1	15	Condition <code>num % 2 == 1</code> prints "Number is Even"	Logical Error	Print "Odd" for odd numbers
2	17	Else block prints "Number is Odd"	Logical Error	Print "Even" for even numbers
3	25	Marks ≥ 90 assigned Grade B	Logical Error	Assign correct highest grade
4	27	Marks ≥ 80 assigned Grade A	Logical Error	Correct grade sequence
5	48	Age > 18 prints "Not Eligible for Voting"	Logical Error	Print "Eligible for Voting"
6	50	Else block prints "Eligible for Voting"	Logical Error	Print "Not Eligible for Voting" when required
7	59	Larger number printed as smaller number	Logical Error	Print actual smaller value
8	61	Else block prints larger value as smaller	Logical Error	Print actual smaller value
9	72	Comparison uses <code><</code> while finding largest number	Logical Error	Compare correctly for largest value
10	75	Comparison uses <code><</code> while finding largest number	Logical Error	Compare correctly for largest value
11	87	Integer division in average calculation	Logical Error	Use floating-point division
12	97	Formula used is circumference-related, not area	Logical Error	Use correct circle area formula
13	109	Bonus percentage 0.02 instead of intended higher rate	Logical Error	Apply correct bonus rate
14	121	Month 1 mapped to February	Logical Error	Map to correct month

S.No.	Line No.	Error	Error Type	Error Correction
15	125	Month 2 mapped to March	Logical Error	Map to correct month
16	129	Month 3 mapped to April	Logical Error	Map to correct month
17	133	Month 4 mapped to May	Logical Error	Map to correct month
18	148	isPrime remains 1 when factor found	Logical Error	Set flag appropriately
19	154	Prime number printed as "Not Prime"	Logical Error	Print correct result
20	156	Non-prime number printed as "Prime"	Logical Error	Print correct result
21	168	Discount stored as fixed value 5 instead of percentage/amount logic	Logical Error	Apply correct discount calculation
22	172	Final amount calculated using addition	Logical Error	Subtract discount from purchase amount
23	183	sum = i overwrites previous sum	Logical Error	Accumulate values properly
24	195	Temperature < 40 prints "Cold Weather"	Logical Error	Use appropriate temperature condition
25	205	Grade A prints "Average Performance"	Logical Error	Print appropriate performance message
26	217	Rectangle area calculated using addition	Logical Error	Use multiplication
27	227	Percentage ≥ 60 prints "Fail"	Logical Error	Print "Pass"
28	229	Else block prints "Pass"	Logical Error	Print "Fail" when required

Problem 7 :

S.No.	Line No.	Error	Error Type	Error Correction
1	9	Loop condition $i \leq 5$ accesses marks[5]	Array Out of Bounds Error	Use $i < 5$
2	24	Integer division used in $\text{average} = \text{total} / 5$	Logical Error	Use type casting for accurate floating-point result
3	32	Loop starts from index 1, leaving numbers[0] uninitialized	Logical Error	Start loop from index 0
4	32	Loop condition $i \leq 10$ accesses numbers[10]	Array Out of Bounds Error	Use valid indices 0 to 9
5	46	String "ComputerScience" exceeds size of name[10]	Buffer Overflow Error	Increase array size or shorten string
6	54	Direct assignment <code>city = "Delhi"</code> to character array	Compilation Error	Use string copy function
7	63	String comparison using <code>==</code>	Logical Error	Use string comparison function
8	76	Use of <code>gets()</code> function	Unsafe Function Error	Use a safer input function with size limit
9	84	<code>strcat()</code> may overflow firstName[20]	Buffer Overflow Error	Ensure destination array has sufficient space
10	94	Character array grade not null terminated before printing as string	Runtime / Logical Error	Add string terminator <code>'\0'</code>
11	102	Negative array index <code>arr[-1]</code>	Array Out of Bounds Error	Use valid index range
12	110	<code>%s</code> with <code>scanf()</code> may overflow department[10]	Buffer Overflow Risk	Limit input size

S.No.	Line No.	Error	Error Type	Error Correction
13	116	String "student@gmail.com" exceeds size of email[15]	Buffer Overflow Error	Increase array size
14	127	Writing to scores[3] when array size is 3	Array Out of Bounds Error	Use valid indices 0 to 2
15	129	Reading scores[3] outside array bounds	Array Out of Bounds Error	Access only valid indices
16	133	String "Programming" exceeds size of subject[10]	Buffer Overflow Error	Increase array size or shorten string
17	143	String comparison username == "admin"	Logical Error	Use string comparison function
18	156	String "987654321012" exceeds size of mobile[10]	Buffer Overflow Error	Increase array size
19	168	Loop condition i <= 5 accesses attendance[5]	Array Out of Bounds Error	Use i < 5
20	178	Accessing course[20] outside valid index range	Array Out of Bounds Error	Use index within array limits
21	190	Character array college not null terminated	Runtime / Logical Error	Add '\0' at the end
22	210	String "IT-A" exceeds size of section[3]	Buffer Overflow Error	Increase array size to store full string and terminator
23	36-40	Printing numbers array may display garbage value at index 0	Logical Error	Initialize all array elements properly
24	56	Assignment to array name is not allowed in C	Compilation Error	Copy string contents instead of assigning
25	118	Buffer overflow causes memory corruption in email array	Runtime Error	Allocate sufficient storage
26	156	Mobile number string requires space for null terminator	Buffer Overflow Error	Allocate enough characters including '\0'

S.No.	Line No.	Error	Error Type	Error Correction
27	84	Concatenated string length may exceed destination capacity	Runtime Error	Check combined length before concatenation
28	143	Username validation will always fail even for matching text	Logical Error	Compare string contents rather than addresses

Problem 8 :

S.No.	Line No.	Error	Error Type	Error Correction
1	15	Function add() called with one argument instead of two	Function Argument Error	Pass both required arguments
2	21	Function average() called with two arguments instead of three	Function Argument Error	Pass all three required arguments
3	61	Function calculateGrade() called without declaration or definition	Compilation / Linker Error	Declare and define the function
4	67	Function circleArea() called without declaration or definition	Compilation / Linker Error	Declare and define the function
5	75	Function sumArray() called without declaration or definition	Compilation / Linker Error	Declare and define the function
6	81	Function displayResult() called without declaration or definition	Compilation / Linker Error	Declare and define the function
7	97	Base case of factorial returns 0 for n == 0	Logical Error	Return 1 for factorial of 0

S.No.	Line No.	Error	Error Type	Error Correction
8	117	maximum() returns smaller value when a > b	Logical Error	Return the larger value
9	119	maximum() returns smaller value in else block	Logical Error	Return the larger value
10	132	Recursive call fibonacci(n) causes infinite recursion	Logical Error	Use smaller arguments in recursive calls
11	145	Integer division in percentage calculation may produce incorrect result	Logical Error	Use proper arithmetic/type casting
12	155	Non-void function simpleInterest() uses return; without value	Compilation Error	Return the calculated value
13	170	Loop condition i < exp gives incorrect result for exponentiation	Logical Error	Ensure multiplication occurs required number of times
14	187	Function reverseNumber() has no return statement	Compilation Warning/Error	Return the reversed number
15	209	isPrime() returns 1 when number is divisible	Logical Error	Return 0 for non-prime numbers
16	212	isPrime() returns 0 when number is prime	Logical Error	Return 1 for prime numbers
17	226	Loop condition i <= size accesses one element beyond array size	Array Out of Bounds Error	Use i < size
18	233	Recursive function has no terminating condition	Runtime Error (Stack Overflow)	Add a valid base condition
19	243	Rectangle area calculated using addition	Logical Error	Use multiplication for area calculation
20	253	cube() returns n * n instead of cube value	Logical Error	Return cube of the number

S.No.	Line No.	Error	Error Type	Error Correction
21	37	swap(x,y) does not swap original variables	Logical Error	Use pointers or return swapped values
22	40	Output after swap remains unchanged	Logical Error	Modify original variables through references/pointers
23	91	Average calculation returns integer division result	Logical Error	Use floating-point division
24	132	Fibonacci implementation recalculates current term recursively	Logical Error	Use correct recursive formula
25	145	Percentage formula may truncate decimal values	Logical Error	Perform floating-point calculation
26	219	updateValue() modifies local copy only	Logical Error	Use pointer parameter to modify caller variable
27	233	Infinite recursion can exhaust program stack	Runtime Error	Add termination condition
28	226	Array printing function may access invalid memory location	Runtime Error	Restrict access to valid array indices

Problem 9 :

S.No.	Line No.	Error	Error Type	Error Correction
1	13	Uninitialized pointer ptr1 dereferenced	Runtime Error	Initialize pointer before dereferencing
2	17	NULL pointer ptr2 dereferenced	Runtime Error	Assign valid memory/address before use

S.No.	Line No.	Error	Error Type	Error Correction
3	25	Pointer increment moves ptr3 beyond intended variable	Logical Error	Avoid accessing memory not belonging to the object
4	27	Dereferencing incremented pointer may access invalid memory	Runtime Error	Access only valid memory locations
5	33	Accessing *(ptr4 + 10) beyond array bounds	Array Out of Bounds Error	Use valid array indices
6	43	Accessing *ptr5 after free(ptr5)	Runtime Error (Use After Free)	Do not access memory after freeing it
7	51	Allocated memory address lost when ptr6 = NULL	Memory Leak	Free memory before assigning NULL
8	57	malloc(5) allocates only 5 bytes, insufficient for 10 integers	Memory Allocation Error	Allocate enough memory using sizeof(int)
9	61	Writing 10 elements into insufficient allocated memory	Buffer Overflow Error	Allocate adequate memory or reduce writes
10	73	Accessing ptr8[3] when only 3 elements allocated	Array Out of Bounds Error	Use valid indices 0 to 2
11	85	Double free of ptr9	Runtime Error	Free allocated memory only once
12	91	Reading uninitialized dynamically allocated memory *ptr10	Undefined Behavior	Initialize memory before reading
13	99	Loop condition i <= 5 writes to numbers[5]	Array Out of Bounds Error	Use i < 5
14	115	pptr = &value; assigns int* to int**	Type Mismatch Error	Assign address of an integer pointer

S.No.	Line No.	Error	Error Type	Error Correction
15	117	Dereferencing invalid double pointer **pptr	Runtime Error	Ensure proper pointer levels and initialization
16	123	Allocated size 5 bytes is too small for "Programming"	Buffer Overflow Error	Allocate sufficient memory for string and null terminator
17	125	strcpy() used without including <string.h>	Compilation Warning/Error	Include required header file
18	135	Incrementing pointer ptr11++ loses original allocated address	Memory Leak	Preserve original pointer
19	137	Dereferencing incremented pointer outside allocated object	Runtime Error	Access only allocated memory
20	141	NULL pointer ptr12 dereferenced in condition	Runtime Error	Check pointer before dereferencing
21	153	Accessing ptr13[100] beyond allocated range	Array Out of Bounds Error	Use indices 0 to 99
22	161	Writing to memory after free(ptr14)	Runtime Error (Use After Free)	Do not access freed memory
23	167	Assigning 2D array to int * causes incompatible pointer type	Type Mismatch Warning	Use appropriate pointer type
24	169	Accessing *(p + 20) outside matrix bounds	Array Out of Bounds Error	Access only valid matrix elements
25	175	First allocated memory lost when ptr15 is reassigned	Memory Leak	Free previous allocation before reassignment
26	185	Swap function swaps local pointer copies instead of values	Logical Error	Swap values pointed to by pointers

S.No.	Line No.	Error	Error Type	Error Correction
27	193	Loop condition $i \leq \text{size}$ accesses one element beyond array size	Array Out of Bounds Error	Use $i < \text{size}$
28	97	Dynamically allocated memory numbers never freed	Memory Leak	Free allocated memory after use
29	145	Memory allocated to ptr13 never freed	Memory Leak	Free memory before program termination
30	175	Final allocation for ptr15 never freed	Memory Leak	Free allocated memory before exit

Problem 10 :

S.No.	Line No.	Error	Error Type	Error Correction
1	12	Missing semicolon after <code>int choice = 1</code>	Syntax Error	Add ; at the end of the statement
2	14	String "90" assigned to integer variable marks	Type Mismatch Error	Assign an integer value instead of a string
3	20	Function <code>add()</code> called with one argument instead of two	Function Argument Error	Pass both required arguments
4	24	Assignment operator (<code>=</code>) used in <code>if(a = b)</code>	Logical Error	Use comparison operator <code>==</code>
5	29	Variable <code>score</code> used without declaration	Semantic Error	Declare and initialize <code>score</code> before use
6	33	Array index 10 exceeds size of array <code>arr[5]</code>	Array Out of Bounds Error	Use valid index range 0-4
7	35	Negative array index <code>arr[-1]</code>	Array Out of Bounds Error	Use non-negative valid index

S.No.	Line No.	Error	Error Type	Error Correction
8	39	String "ProgrammingLanguage" exceeds size of name[10]	Buffer Overflow Error	Increase array size or shorten string
9	45	Direct assignment city = "Delhi" to character array	Compilation Error	Use string copy function
10	52	String comparison using ==	Logical Error	Compare string contents using string comparison function
11	60	Division by zero (x/y, where y = 0)	Runtime Error	Ensure divisor is not zero
12	64	NULL pointer ptr1 dereferenced	Runtime Error	Assign valid memory before dereferencing
13	68	Uninitialized pointer ptr2 dereferenced	Runtime Error	Initialize pointer before use
14	78	Accessing *ptr3 after free(ptr3)	Runtime Error (Use After Free)	Do not access memory after freeing it
15	80	Double free of ptr3	Runtime Error	Free allocated memory only once
16	84	malloc(5) allocates insufficient memory for 10 integers	Memory Allocation Error	Allocate memory using required size
17	88	Writing 10 elements into insufficient allocated memory	Buffer Overflow Error	Allocate enough memory or reduce loop range
18	94	Memory allocated to ptr5 lost due to reassignment	Memory Leak	Free previous allocation before reassignment
19	100	Function average() called with two arguments instead of three	Function Argument Error	Pass all required arguments
20	108	Age greater than 18 prints "Not Eligible"	Logical Error	Print appropriate eligibility message

S.No.	Line No.	Error	Error Type	Error Correction
21	120	isPrime remains 1 even when divisor is found	Logical Error	Set isPrime = 0 when number is divisible
22	126	Prime number printed as "Not Prime"	Logical Error	Print correct result based on isPrime value
23	130	Function displayReport() declared but not defined	Linker Error	Provide function definition
24	132	Function calculateSalary() called without declaration	Compilation Error	Declare function prototype before use
25	132	Function calculateSalary() not defined	Linker Error	Provide function definition
26	140	fscanf() used without checking whether file opened successfully	Runtime Error	Verify file pointer is not NULL
27	148	Missing colon (:) after case 1	Syntax Error	Add : after case label
28	159	Multi-character constant 'AB' assigned to char	Semantic Error	Use a single character constant
29	167	Writing to scores[3] when array size is 3	Array Out of Bounds Error	Use valid indices 0-2
30	171	Loop condition $i \leq 3$ accesses scores[3]	Array Out of Bounds Error	Use $i < 3$
31	181	Formula for circle area is incorrect ($2 \times \pi \times r^2$)	Logical Error	Use correct area formula
32	185	Semicolon after while(choice > 0) creates empty loop	Logical Error	Remove unnecessary semicolon
33	190	continue used outside loop	Syntax Error	Use continue only inside loops
34	201	Average function performs integer division	Logical Error	Use floating-point division for accurate average

S.No.	Line No.	Error	Error Type	Error Correction
35	138	File abc.txt may not exist, causing invalid file operations	Runtime Error	Check file opening status before reading
36	152	Missing break after case 2	Logical Error	Add break if fall-through is not intended